

BEOPS — pre-paper for the conference: what exists, where the literature stands, what the paper could be

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paper/PRED_RAD_SR_V2_2026-09-06.md as the discussion entry for the paper; does not supersede the working document v1.1, the registry or any evidence · Venue in view: *Creating sustainable commUNiTy* — 6th International Conference on Sustainable Environment and Technologies, University Union – Nikola Tesla, 25 September 2026, online (≤ 6 pages A4, abstract ≤ 200 words, APA); the full-manuscript deadline of 20 August has passed and a late-submission question has not been sent.

Every number in this document was read from a file on the university machine on the morning of 9 September 2026 or from a page that was actually opened; unverified items are marked. Section 1 is in Serbian; the rest is in English because the paper and the public site are English-first.

1 • Sažetak

BEOPS (Belgrade Evidence Observatory) je opservatorija grada zasnovana na dokazima: sistem koji sluša ono što Beograd javno i zakonito objavljuje o sebi, beleži *kada* je nešto stiglo, *kada* je izmereno i *kada* je izvor ćutao, i ne izmišlja ništa što nije primljeno. Na jutro 9. septembra 2026. postoji: registar od 191 izvora sa 154 uhvaćena dokaza dozvole; kolektor koji svakih pet minuta prima podatke sa 32 zvanične stanice kvaliteta vazduha (SEPA), 52 građanska senzora (Sensor.Community) i 27 parkirališta, plus šest novinskih RSS izvora (samo naslov, veza i vreme); javni sajt koji se sam obnavlja svakih 30 minuta; dva organa na lokalnim modelima — razvrstavač vesti i **um**: tri entiteta (Posmatrač, Sumnjalo, Povezivač) koji nad numerisanim činjenicama razmišljaju naglas, odgovaraju jedno drugom i čije svaku rečenicu program proverava pre nego što je iko vidi. Prva noć uma dala je prvi izmereni rezultat: tri entiteta na istoj porodici modela su se do jutra slile u jednu rečenicu — eho koji literatura predviđa — i to je sada zabranjeno kodom, ne molbom.

Ovaj pretpaper kaže šta imamo, gde stoji literatura 2024–2026 po šest tema, koje četiri praznine BEOPS može pošteno da popuni (semantika nepoznatog vremena merenja po opservaciji; odsustvo kao prostorno agregiran zapis prvog reda sa vizuelnom gramatikom; starost podataka na javnom ekranu; um od malih lokalnih modela koji ne sme da izgovori broj kog nema u dokazima — i čiji se iskazi mere), i predlaže tri moguće strukture rada, sa preporukom koji rad ide na ovu konferenciju a koji čeka seriju od nedelju dana.

2 • What exists on 9 September 2026 — the verified inventory

What	State	Where it is read from
Source registry	191 records ; statuses include <code>collected</code> , <code>probe_ok</code> , <code>opted_out</code> , <code>no_coverage</code> , <code>dead</code> , <code>account_required</code> — an absence or a refusal is a record, not a silence	<code>research/SOURCE_REGISTRY.json</code>
Permission evidence	154 access checks passed, 8 forbidden (named refusals honoured), 25 undocumented; every capture is bytes on disk with SHA-256 (robots.txt, headers incl. <code>X-Robots-Tag/Content-Signal/TDM-Reservation</code> , licence page)	<code>research/08-provenance/INDEX.md</code> , <code>LEDGER.jsonl</code> , <code>research/evidence/legal/</code> (local)
Live collector	<code>tools/collect_daemon.py</code> , <code>stdlib</code> , every 5 min (task <code>Beops_Collect</code> , verified at the scheduler): S146 SEPA HVD hourly means for 32 Belgrade stations (interval published), S04 Sensor.Community (~52 sensors within 25 km), S10 Parking Servis (27 lots, no measurement time published), six RSS feeds headline-only; three timestamps per row; missing = <code>null</code> ; seen-set dedupe; receipts hard-linked	<code>research/COLLECTORS.json</code> , <code>data/live/</code> (local)
Corrections	13 (C-001...C-013), appended never edited — including the source that labels local time as UTC, the index that called permitted sources refusals, the organ that bound answers by position	<code>research/08-provenance/CORRECTIONS.md</code>
Organ 1: news-sorter	0.1.1, local model, categories + zones from a fixed gazetteer, binds by echoed headline (C-013); ~120 headlines sorted by the morning	<code>tools/organ_news.py</code> , <code>research/ORGANS.json</code>
Organ 2: mind	0.3.4 (\$4.4): three temperaments on local models, organelles, English thinking + Serbian voice, validator, notebooks, claims scored; drip every 4 min (<code>Beops_Mind</code>); first night: 76 drops, 37 thoughts accepted, 23 refused, 4 claims (1 true, 2 false, 1 open), 16 organelle drops	<code>tools/organ_mind.py</code> , <code>data/live/derived/mind/</code> (local), <code>research/03-models/MIND_ARCHITECTURES_2026-09-09.md</code>
Base map	Natural Earth 1:10m (public domain, S98) after five refused routes (GeoSrbija 401, Nominatim robots, ohsome 403, Overpass robots, Geofabrik robots) — all recorded, none circumvented; own <code>stdlib</code> shapefile reader	<code>tools/fetch_basemap.py</code> , <code>public/basemap-belgrade.json</code>
Static context	Kontur population (S120, CC BY 4.0, HDX release 2022-06-30): 2 452 H3 hexagons, 1.72 M people in the window; register of further lawful layers with status	<code>research/STATIC_LAYERS.json</code> , <code>tools/fetch_static.py</code> , <code>public/context-population.json</code>
Public site	https://3esign.github.io/beops/ — generated by <code>tools/build_site.py</code> from the registry, the index, the corrections and the last snapshot; republished every 30 min (<code>Beops_Publish</code>); MONOLOG + PULS (bilingual; map pulses only on a real reception; population hexagons; the	<code>docs/</code> , <code>tools/publish_github.ps1</code> , <code>github.com/3esign/beops</code> (export without evidence; MIT code / CC BY 4.0 documents)

What	State	Where it is read from
	mind's drops at the minute they fell), TRAKA, four generated maps, the PDFs	
Documents	Working document v1 (33 pp) and v1.1 (36 pp, three generated maps)	research/06-paper/
Tests	98 offline tests green (collector, organs, base map, mind), run before every deployment; every deployed file's hash verified on the machine	research/test_*.py, npm test

What does **not** exist, said plainly: a time series longer than one day (collection began 20:52 UTC on 8 September); any evaluation of the mind beyond one night; municipality boundaries on the map (every lawful route refused); a domain (GitHub Pages is the address); a submitted manuscript.

3 • Where the literature stands (2024–2026) — six themes, and where BEOPS sits

The references below were verified on 8–9 September 2026 (arXiv, Crossref, publisher pages) unless marked $\triangle$; the full annotated list with links is `research/_trail/SPOLJNI_PREGLED_2026-09-08_ISTRAZIVANJE.md` (Report C).

3.1 Urban sensing and urban LLM agents

- Hou, Zhang, Li et al. (2025), *Urban sensing in the era of large language models*, The Innovation 6(1), 100749 — the programmatic statement: LLMs as commonsense reasoners over urban data; limits named as multimodal fusion, spatio-temporal cognition, cultural adaptability, privacy. Nothing on absence, provenance or what a model may *not* say.
- HKUST survey (2025), *Large Language Model Powered Intelligent Urban Agents: Concepts, Capabilities, and Applications*, arXiv:2507.00914 — taxonomy by domain (planning, transport, environment, safety, society) and by workflow (sensing, memory, reasoning, execution, learning); "trustworthiness and evaluation" named as open; small/local models absent from the abstract.
- Ye et al. (2026), *Towards Agentic Urban Digital Twins (AUDiTs)*, Urban Informatics — four agent roles (data curator, scenario planner, dialogue mediator, ethics & safety), human oversight "central rather than peripheral", uncertainty via calibration and intervals; hallucination and drift named as risks, not solved.
- UrbanInsight (2025), arXiv:2509.00936 — LLM-generated filtering rules at the edge for smart-city twins; the edge framing is the nearest to BEOPS's under-a-desk constraint, the evaluation not readable from the abstract.
- Salemi et al. (2025/26), *LLM-Based Multi-Agent Blackboard System for Information Discovery*, arXiv:2510.01285; Li et al. (2025), *Open Dataset Search in the LLM Era*, arXiv:2509.00728; Kreutz et al. (2025), arXiv:2507.04444 ("a tool rather than a partner").

Where BEOPS sits: the surveys describe agents that *act* in cities; BEOPS's mind is an agent that may only *think about receipts*, under a validator that refuses any number not in the evidence. That inversion — evidence bounds the language, not the other way round — is not in this literature.

3.2 Multi-agent talk: councils, debate, echo

- *Council Mode* (arXiv:2604.02923, April 2026): three heterogeneous frontier models in structured consensus — 35.9 % relative reduction on HaluEval, 7.8 points on TruthfulQA, at 2–3× latency; a misconception shared by all experts is reproduced, not caught.
- The hallucination-amplification literature on multi-agent debate: echo, sycophancy, confident-but-wrong voting and context bloat make debate *worse*; the mitigations named are evidence grounding, independent verification, heterogeneity, provenance logging and a human interrupt.

Where BEOPS sits: the first night of the drip reproduced the amplification result in a system of three 1.5–3B models within eight hours (§4.4) — and then made the mitigations code: only validated utterances are exchanged, restating is refused by a 4-gram overlap test, the notebook reads back refusals rather than accepted text, a "new since you last spoke" fact steers attention. A measured, repaired echo collapse at small scale is a result the council papers (frontier models, benchmark questions) do not report.

3.3 Time semantics: what the standards can and cannot say

- OGC/ISO 19156:2023 (OMS, OGC 20-082r4): `phenomenonTime` exactly one, `resultTime` exactly one — **no way to say "phenomenonTime unknown"**.
- OGC SensorThings API 2.0 draft (OGC 23-019, public comment closed January 2026): `resultTime` becomes optional — *omitted*, not null — the silent omission BEOPS argues against.
- W3C SSN 2023 Edition (WD, latest 7 Sept 2026) with the OMS module; W3C Time Ontology (CR 2022) has nominal positions but no explicit "unknown" instant; PROV-O (2013) has no successor; DCAT 3 (2024) adds `DatasetSeries` and `wasGeneratedBy`; Frictionless Data Package v2 (2024) promotes per-field `missingValues`.
- Akidau et al. (2021), watermarks in stream processing — event time vs processing time; Maldonado et al. (2026), arXiv:2604.01440, intentional out-of-order streams; Wang, Zheng, Zhang (2026), *StreamReason-Bench*, arXiv:2608.12348 — no LLM exceeds 34 % exact match on event-time semantics.

Where BEOPS sits: three timestamps per row (measured / published / received), `phenomenonTimeUnknown` as an explicit boolean, `phenomenonTimeCorrected` as a marked *estimate* when a source mislabels its clock (SEPA labels CEST as Z — C-013's neighbour). The claim is an **OMS/SOSA extension profile**, not a new model.

3.4 Absence, staleness and their visualization

- Alsufyani, Forshaw, Johansson Fernstad (2024), *Visualization of missing data: a state-of-the-art survey*, arXiv:2410.03712 — tabular, not spatio-temporal sensor networks; Johansson Fernstad et al. (2025), *To Measure What Isn't There*, arXiv:2505.23447; Ruddle (2026), PLOS One 21(8); Song & Szafir (2018), *Where's My Data?*, IEEE VIS —

how readers misjudge charts with missing points; Tennant & Randall (2025), IEEE TVCG — temporal uncertainty is not a category; Wang et al. (2021), *Journal of Visualization* — no "unknown timestamp" type; Kitchin, *Urban data and city dashboards: six key issues*.

- △ Not found: any 2024–2026 controlled study of *data age / staleness* on public dashboards; only vendor "freshness checks".
- Ferrer-Cid et al. (2025), JNCA — the mainstream treats gaps as imputation targets.

Where BEOPS sits: absence is counted, not drawn — a quorum row of filled/hollow cells per source per cadence slot; "silence is a record, not a zero" is a sentence in the monologue; no imputation anywhere by rule. Map 3 of the working document is the first *map of what the observatory cannot hear* for Belgrade.

3.5 Observatories, twins, and Belgrade

- Wolf, Stiles, Miller, Dawson, Mills, Blythe, Morley (2025), *Building enduring smart city data platforms*, Frontiers in Sustainable Cities 7 — the UK Urban Observatories and Smart Columbus as the concept paper; Newcastle's Urban Observatory as the closest living analogue (Report A).
- Dawkins & Kitchin (2025), *Urban digital twins: digital twins for participatory steering*, NM&S 27(8); Moroni (2025), *Insurmountable limitations of city-scale digital twins?*, Comput. Urban Sci. 5:17; Hiller et al. (2026), *Mapping Urban Digital Twins Across Regions*, Smart Cities 9(3) — 99 twins, over half not operational, **no Serbian or Balkan case**; Azadi et al. (2025), Smart Cities 8(1).
- Serbia: Stojanović et al. (2024), Sensors 24(13) — a 15-municipality low-cost PM network with a 75 % coverage rule and no published outage statistics; Ramadani et al. (2025), Atmosphere 16(9) — low-cost monitors in Belgrade schools; Arnaut et al. (2025), Urban Science 9(10) — Belgrade accessibility on open data; data.europa.eu (2025) Serbia factsheet — the portal "does not allow users to view unavailable datasets"; CINS (2020) on the reliability of official air-quality data (journalism; △ no 2024–26 academic update).

Where BEOPS sits: an observatory, explicitly not a twin, positioned as an instance of Dawkins & Kitchin's "steering representation"; the first Belgrade-wide, dated inventory of what publishes, what refuses and what is dark (191 records).

3.6 Law: text-and-data mining, opt-outs, the AI Act

- Serbia's draft copyright act (consultation 16 March – 6 April 2026) transposes 2012/28, 2014/26 and Marrakesh — **no TDM exception, no mention of Directive 2019/790** (three sources; △ the draft text itself not opened); Copyright Act Art. 6(2) excludes official materials of public bodies; ZZPL Art. 88 is the scientific-expression exemption.
- Quintais (2025), CLSR 56; Senftleben (2026), IIC 57(1); Teilmann-Lock & Savin (2025), CLSR 58; CJEU C-250/25 *Like Company v Google* (AG opinion scheduled 3 Sept 2026 — △ check before submission); Longpre et al. (2024), *Consent in Crisis*, NeurIPS D&B — the AI data commons closing.
- Machine-readable preferences: IETF AIPREF `draft-ietf-aipref-vocab-07` (Aug 2026, has an "unknown" state); Cloudflare Content Signals (Sept 2025); `ai.txt` draft (June 2026)

— norm fragmentation. EU AI Act Art. 50 transparency; algorithm registers (Amsterdam, Helsinki).

Where BEOPS sits: permission is a first-class field captured as bytes with a three-state verdict in which unknown is never permission; named opt-outs honoured; headlines never republished as a corpus; every model utterance machine-marked; an algorithm register with an editor of record; model licences recorded per model (and one of them — Qwen2.5-3B's research licence — already flagged as a constraint on any product).

4 • The contributions BEOPS can claim honestly

Four, scoped so that a reviewer who checks the files finds exactly what the sentence says.

1. **A per-observation semantics for unknown measurement time**, as an OMS/SOSA extension profile: `phenomenonTimeUnknown`, three timestamps, marked estimates for mislabelled clocks — with a rendering grammar (a stroke says where the value came from, a ground says how well its time is known; only observation is chromatic). Precedent runs out here (§3.3-3.4).
2. **Absence as a spatially aggregated first-class record** with a visual grammar (quorum cells, silence sentences, a map of what cannot be heard), and no imputation by rule — a deliberate counter-position to the mainstream (§3.4).
3. **A verification layer for agent-discovered web sources** — permission captured as evidence, three-state verdicts, refusals as records — novel in application, not mechanism (§3.6).
4. **A mind of small local models bounded by evidence**: numbers only from a digest, citations bound by content, predictions only as scorable claims, refused utterances logged and never echoed; measured — including its first failure (echo collapse) and its repair (§4.4).

Not claimed: forecast-vs-measurement (OMS `validTime`, DQV cover it — BEOPS contributes enforcement and UI); PROV-O for sensors (standard practice); the critique of twins (saturated); LLM data discovery (two surveys and an architecture exist); the Serbian TDM gap as analysis (Quintais, Senftleben cover it — BEOPS documents the fact).

4.4 The mind's first night, as data

76 drops between 00:50 and 08:38 UTC. Accepted thoughts 37, refused 23 — 29 of the refusal reasons were "a number not in the digest", 4 "hypothesis stated as fact", 2 "cites nothing inside the text". Serbian renderings 17, refused 15 (the voice changed citations — fixed by stripping them before rendering), failed 5 (timeouts). Claims 4: a reception claim true, two spread claims false, one open. Organelle drops 16 (the surprise-ranker on llama3.2:1b; the embedding linker joined in the morning — on real Serbian headlines the same event scored cosine 0.725 and a different one 0.266, so the threshold is 0.70, set after measuring, not before). And the finding: by morning the Observer, the Skeptic and the Connector were saying one sentence — the same three mechanisms the debate literature names (echo of the others, re-reading one's own accepted text, a slowly changing world) produced convergence in eight hours on 1.5-3B models. Version 0.3.4

refuses restatement by code. Whether that holds over a week is the experiment now running.

5 • What the paper could be — three structures

Option A — "An evidence-first observatory for Belgrade: what the city publishes, when, and where it goes quiet" (recommended for 25 September)

Fits six pages; every number exists today; the contribution is the *record* and the *grammar*, which do not need a week of series.

§	Content	Pages	Figure
1	Problem: dashboards and twins present what they have as if it were the city; the questions <i>when was this measured</i> and <i>what is missing</i> have no place in them (Kitchin; Song & Szafir; Hiller et al.)	0.6	—
2	Doctrine: five states; received $\neq$ measured; missing $\neq$ zero; forecast $\neq$ measurement; every claim carries source, time, space, unit, permission	0.6	the state alphabet (stroke $\times$ ground)
3	Method: registry with refusals and absences as records; permission as captured bytes with three-state verdicts; the collector's three timestamps; corrections appended	1.0	the permission index table (154 / 8 / 25)
4	The Belgrade inventory: 191 records — what publishes with measurement time (SEPA), what publishes without (parking), what refuses (named), what is dark (traffic counters, AVL, seismic archive); the base-map refusals	1.2	Map 1 (instruments) and Map 2 (coverage)
5	The grammar of absence and time: quorum cells, silence sentences, the untimed state, the corrected clock (SEPA labels local time as Z)	1.0	Map 3 (last 24 h) + a TRAKA strip
6	Position and limits: extension profile not new model; one day of series; no municipality boundaries; the legal frame (no TDM exception; Art. 6(2)); what the mind is and is not (one paragraph, pointing to Option B)	0.8	—
7	Conclusion + the public artefact (site, repository, index)	0.3	—

Abstract $\leq$ 200 words; APA references $\approx$ 18 (the ones in §3 marked verified).

Option B — "A mind bounded by evidence: small local models thinking aloud about a city, and what refusing them teaches" (for a later venue, after a week of drips)

The architecture (§4.4), the validator as the contribution, the measured echo collapse and its repair, refusal rates by model and temperament, claim accuracy after a week, the voice as a measured bilingual step; positioned against Council Mode and the amplification literature. Needs: seven days of scoreboard, the editor's weekly sample, one ablation (notebook on/off, round-2 on/off) — all supported by the files without new code.

Option C — "Unknown time, counted absence: a rendering grammar for observations whose time nobody knows" (visualization venue)

The narrowest and the most original: the fifth state, the quorum bar, the untimed colour, the corrected clock — with a small user study (§12.5 of the working document) that does

not yet exist. Needs the study; not before spring.

Recommendation: A for the conference, because it is true today and defensible on files; B as the second paper from the same repository once the scoreboard has a week; C when the user study is designed. One contribution per paper, as the project rule says.

6 • What must be true before submission (whichever option)

- The late-submission question sent and answered (the deadline has passed; a member of the scientific committee is a co-author).
- The manuscript's every number re-read from the files on the day of submission; the working document's honest-verdict discipline kept in the paper's own last paragraph.
- The AG opinion in C-250/25 checked (scheduled 3 September 2026).
- The Serbian draft copyright act's actual text opened, not only the three reports about it.
- Figures generated by `make_maps.py` from the snapshot of the submission day, so the captions and the maps carry the same timestamp.
- No sentence about the mind that the scoreboard does not support.

Honest verdict

Checked: the inventory in §2 against the files on the machine on the morning of 9 September (registry, index, receipts, the mind's rows read one by one, the scheduler); the references in §3 against the annotated trail file where each carries a verified/unverified mark; the first-night numbers against `data/live/derived/mind/` directly. **Concluded:** BEOPS can defend four scoped contributions today and one of them (the mind) already has a measured negative result worth publishing; Option A is the paper that is true now. **Not looked at:** whether the conference accepts late manuscripts; the full text of Batty (2025) and Raca et al. (2022); the AG opinion; the Serbian draft law's text; anything the drip produces after 08:38 UTC on 9 September.